

Ignoring the Decoy: Exposing and Tackling Forensic Distractions in Image Forgery Localization using Masked Convolutions

Xander Staelens, Peter Lambert, Glenn Van Wallendael, and Hannes Mareen
IDLab, Ghent University - imec, Ghent, Belgium

firstname.lastname@ugent.be

Abstract

Manipulating images has become increasingly easier due to advanced editing tools, leading to a surge in forged images being shared online, raising the need for robust multimedia forensics. Particularly, image forgery localization (IFL) methods aim to detect and localize image tampering. In this paper, we expose and quantify the impact of forensic distractions on state-of-the-art IFL models. We define distractions as benign visual elements that occur in real-world scenarios, such as logos and visible watermarks. Our analysis, first of its kind, reveals that some recent IFL models (like CAT-Net and TruFor) are highly sensitive to distractions, suffering average performance degradations of 15.06% and 55.65%, respectively. To address this issue, we propose masked convolutions, which enable CNN-based IFL methods to ignore masked regions during inference. Masked convolutions require no retraining, allowing existing models to be adapted seamlessly. Distractions can be manually masked, or automatically through a two-step detection process. Experiments show that our proposed approaches significantly improve robustness against distractions, reducing performance degradation to just 4.15% for CAT-Net and 2.94% for TruFor. This demonstrates our method's potential as a distraction-aware technique to enhance the real-world applicability of IFL models.

1. Introduction

Due to the growing sophistication and accessibility of image manipulation tools (often based on Generative Artificial Intelligence), creating high-quality manipulated images is no longer limited to skilled professionals [20]. However, these tools are often exploited for malicious purposes, such as fabricating fake news, spreading misinformation, impersonating individuals, and facilitating fraud. Hence, the need for robust and reliable multimedia forensics tools has become increasingly important.

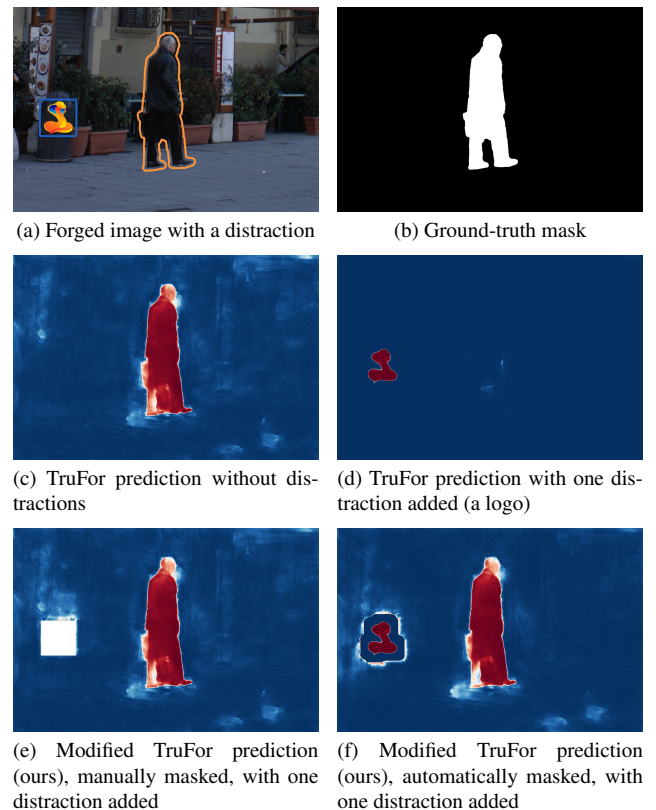


Figure 1. Before adding the logo as a forensic distraction (in blue), TruFor [4] detects the forgery (in orange) with high accuracy (c). However, after adding the distraction and applying lossy JPEG compression, TruFor fails to detect the forgery (d), as the distraction causes it to lose focus. In contrast, TruFor adapted with our proposed masked-convolution approach is able to ignore the distraction and regain focus on the forgery, both using manual masking (e) and automatic masking (f). In (e), the white box indicates the masked region. Note: the same compression was applied to the non-distracted image (c) before making the TruFor prediction.

Despite the significant progress in the field of multimedia forensics, current state-of-the-art models still face several critical challenges, limiting their effectiveness in real-world applications. One core issue is the limited robustness against post-processing operations, such as compression and resizing. Another common post-processing operation is the addition of benign visual elements, such as captions, visible watermarks, and logos. These superficial, often non-malicious modifications, in this paper referred to as *forensic distractions*, can significantly impact the performance of image forgery localization (IFL) models. Forensic distractions sometimes result in a full loss of focus, causing models to entirely ignore present forgeries. Fig. 1 illustrates the impact of forensic distractions on the performance of TruFor [4], a state-of-the-art IFL model. Although forensic distractions frequently occur in real-world images, their effect has never been formally studied.

The aim of this research is to investigate and quantify the impact of forensic distractions on IFL models and propose a solution to mitigate this problem. This paper’s contributions are: (i) Quantifying the impact of forensic distractions on IFL performance, in Section 3. (ii) Proposing the novel masked convolutions that replace standard convolutions in existing convolutional neural network (CNN) based detection models, allowing distraction regions to be ignored without requiring any retraining (i.e., model weights can be reused), in Section 4. (iii) Introducing a two-step detection process with thresholding-based mask generation to achieve automatic distraction-aware IFL, in Section 4.3. (iv) Evaluating the proposed approach by integrating it into TruFor and CAT-Net and demonstrating its potential to significantly improve robustness against distractions, in Section 5.

2. Related Work

2.1. Image Forgery Localization

Manipulated images contain both authentic and forged regions. IFL models aim to localize these forgeries, and produce a pixel-level anomaly map, assigning each pixel a likelihood of being manipulated. This is done by analyzing subtle forensic traces, such as camera noise [1] and compression artifacts [14], introduced by the manipulation process.

IFL models fall into two main categories: model-based and data-driven deep learning methods. Model-based methods rely on handcrafted features and assumptions about the statistical or physical properties of authentic images [6, 10, 11, 23]. Their effectiveness often depends on how well these assumptions hold, limiting generalization in real-world settings.

Data-driven methods (like TruFor [4], used in Fig. 1) learn to detect and localize forgeries directly from data, for example using CNNs to capture both low-level forensic traces and high-level semantic inconsistencies [4, 7, 13].

These black-box methods often outperform model-based methods but offer limited interpretability.

A key challenge in IFL is providing robustness against post-processing, such as compression [16], resizing, and forensic distractions (the latter elaborated in Section 3).

2.2. Restricting Attention to Relevant Regions

As IFL models can be sensitive to forensic distractions (as demonstrated in Section 3), we desire methods that allow models to restrict attention to relevant regions while ignoring known distractions. This requires a mechanism to control the spatial flow of information through the network. We therefore review related work that attempts to guide or restrict attention in CNNs by enhancing relevant or suppressing irrelevant regions.

Attention mechanisms help CNNs focus on relevant image regions [5, 22], while reverse attention [8] aims to suppress less relevant ones. However, neither approach uses a distraction mask. Instead, the model must learn which regions to focus on or ignore, which is a difficult task in itself. Additionally, (reverse) attention only slightly reduces the influence of ignored regions, often adjusting values toward zero but not fully ignoring them. This makes such methods ill-suited for the task of distraction-aware forgery localization, where strict exclusion is required. Moreover, attention mechanisms must be integrated into the architecture during training and cannot be easily added to existing models post hoc.

Liu *et al.* [12] proposed a more explicit masking mechanism for image inpainting, using partial convolutions guided by a binary mask. The convolution operates only on valid pixels (mask value 1), with outputs scaled based on the number of valid inputs in the kernel window. If no valid pixels are present, the output is zero. The mask is updated after each layer, gradually expanding valid regions (i.e., shrinking the mask) as inpainting progresses.

While our method (presented in Section 4) is conceptually related, Liu *et al.*’s approach is not directly applicable to distraction-aware forgery localization for two reasons. First, the renormalization step used in partial convolutions is tailored for content generation rather than feature extraction. In IFL, preserving the relative magnitudes and spatial consistency of forensic traces is critical; rescaling the outputs would distort these signals. Second, their method assumes all regions will eventually be filled in, while in our setting distraction regions must remain excluded throughout the entire network. Therefore, our approach enforces strict, non-adaptive exclusion of distractions without renormalization or mask shrinking.

3. Motivation: Forensic Distractions

In this section, we evaluate the impact of forensic distractions on state-of-the-art IFL models, aiming to quantify the

decrease in detection performance caused by their presence.

As images circulate online, they are often modified with additional visual elements — either intentionally (e.g., logos, watermarks, captions) or unintentionally (e.g., notification bars in screenshots). We refer to these elements as forensic distractions. While they do not change the image’s semantic content or introduce fake visual information, IFL models may misinterpret them as manipulations and prioritize them over the actual forgery.

3.1. Experimental Setup

Forensic methods We evaluate seven IFL models: four model-based methods (DCT [23], ADQ1 [11], BLK [10], and CAGI [6]), and three data-driven methods (TruFor [4], CAT-Net [7], and PSCC-Net [13]). These methods were chosen for their wide adoption in forensic dashboards and libraries, making them representative of state-of-the-art approaches used in practice [15, 18, 19, 21].

Datasets Four datasets are used: DSO-1 [2] (100 images), VIPP [3] (62 images), OpenForensics [9], and IMD2020 [17] (200 images each, randomly selected). All images are resized to 1536×1024 pixels.

Distractions Each image is distracted with a randomly selected logo sourced from freepik.com. The logo is placed at a random position and scaled to a random size, ranging between 4% and 24% of the image width. Note that the added distraction can overlap with the present forgery.

Compression All images are JPEG-compressed after adding distractions with a random compression factor between 95 and 85. Without compression, we noticed that IFL models are significantly less affected, often simply detecting both the distraction and the forgery. Lossy compression (even at a high quality) hides subtle forensic traces, leading to a loss of focus, as illustrated in Fig. 1. To ensure a fair comparison, each non-distracted image underwent the same compression as the corresponding distracted image.

Detection score To evaluate the impact of distractions, we compare the model predictions (P) to the ground truth (G) using an *adapted* F1* score (1), which excludes distraction pixels marked by a binary mask (M). In other words, only the pixels corresponding to $M = 0$ are considered. This ensures models are neither rewarded nor penalized for detecting the distraction.

$$\text{F1* score}(P, G, M) = \frac{2 \text{TP}_{M=0}}{2 \text{TP}_{M=0} + \text{FP}_{M=0} + \text{FN}_{M=0}} \quad (1)$$

To evaluate the impact of forensic distractions on IFL models, we calculate the F1* scores on both the original and distracted forged images. Note that the same compression is applied to both images.

3.2. Results

The top part of Table 1 shows the mean F1* scores of the evaluated IFL methods on the original and distracted im-

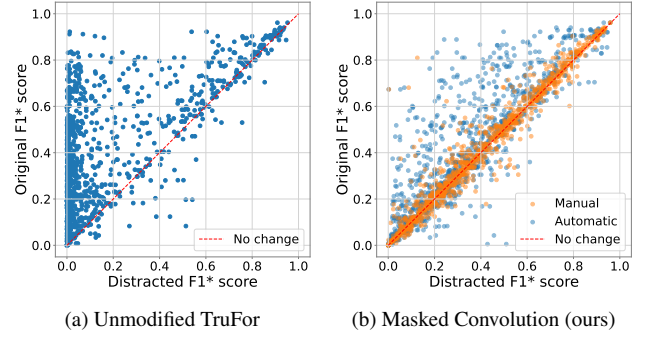


Figure 2. Scatter plots of the F1* scores (1) of unmodified TruFor [4] (a) vs. TruFor with the proposed masked-convolution approach (b), both with manual and automatic masking. Points farther to the left of the diagonal indicate stronger performance degradation due to forensic distractions. While the unmodified TruFor shows many points to the left, the adapted TruFor (both manual and automatic) shows a clear shift toward the diagonal, reflecting improved resilience to distractions. Notably, the points for manual masking (orange) cluster closely around the diagonal, indicating very high resilience to distractions.

ages, and the corresponding decreases, for each dataset. We describe the observations below.

Model-based methods show strong resilience against forensic distractions, with an average performance drop of just 0.78%. Most notably, ADQ1 appears completely unaffected by distractions, while CAGI even performs slightly better with distractions present.

PSCC-Net scores poorly on all datasets, performing even worse than the model-based methods. Its limited performance further decreases with the addition of distractions, resulting in an average drop of 16.64%, indicating moderate sensitivity to distractions.

CAT-Net ranks second on average for non-distracted images, but also shows vulnerability to distractions, with a performance drop of 15.06%. Fig. 3 shows two manipulated images with their CAT-Net predictions with and without distractions present. The first image shows a large performance drop due to the distraction, while the second shows a loss of focus on one of the two present forgeries.

TruFor achieves the highest accuracy on non-distracted images, outperforming all evaluated IFL models with an average F1* score of 0.399. However, it exhibits a large sensitivity to distractions, with performance dropping by 55.65% to 0.177. To better understand this vulnerability Fig. 2a shows TruFor’s performance on each image across all datasets, comparing F1* scores with (x-axis) and without (y-axis) distractions. Points to the left of the diagonal indicate degraded performance, with a cluster on the far left representing cases where detection completely fails. In contrast, the cluster in the upper right shows images where the forgery remains easily detectable despite the presence of

Table 1. Performance of IFL models before (orig) and after (distr) adding forensic distractions, as well as their difference (diff), measured in F1* score (1). The best performance on distracted images for each dataset is highlighted in **bold**, and the second-best is underlined. Most notably, TruFor shows a significant decrease in performance, addressed by the proposed Mod. TruFor.

Model	DSO-1		VIPP		OpenForensics		IMD2020 ¹		Average	
	orig	distr (& diff)	orig	distr (& diff)	orig	distr (& diff)	orig	distr (& diff)	orig	distr (& diff)
DCT [23]	.185	.183 (-0.81%)	.150	.139 (-7.73%)	.120	.115 (-4.34%)	.131	.124 (-5.44%)	.147	.140 (-4.34%)
ADQ1 [11]	.214	.215 (0.64%)	.176	.177 (0.92%)	.113	.112 (-1.28%)	.133	.134 (0.42%)	.159	.159 (0.33%)
BLK [10]	.181	.178 (-2.02%)	.137	.130 (-5.08%)	.080	.076 (-4.19%)	.115	.112 (-2.57%)	.128	.124 (-3.29%)
CAGI [6]	.224	.228 (1.67%)	.158	.166 (4.96%)	.065	.068 (4.74%)	.111	.120 (7.90%)	.139	.145 (4.20%)
PSCC-Net [13]	.061	.051 (-16.91%)	.135	.120 (-11.15%)	.078	.058 (-26.22%)	.098	.082 (-16.33%)	.093	.078 (-16.64%)
CAT-Net [7]	.073	.065 (-11.42%)	.136	.093 (-31.66%)	.281	.230 (-18.20%)	.409	.376 (-8.04%)	.225	.191 (-15.06%)
TruFor [4]	.422	.225 (-46.53%)	.321	.103 (-67.96%)	.405	.127 (-68.56%)	.447	.252 (-43.72%)	.399	.177 (-55.65%)
Mod. CAT-Net (ours)										
Manual	.073	.068 (-6.76%)	.136	.138 (1.92%)	.281	.260 (-7.62%)	.409	<u>.395</u> (-3.31%)	.225	.215 (-4.15%)
Automatic	.075	.068 (-9.23%)	.144	.145 (0.71%)	.276	.248 (-10.21%)	.401	.377 (-5.95%)	.224	.209 (-6.46%)
Mod. TruFor (ours)										
Manual	.422	.406 (-3.70%)	.321	.308 (-4.04%)	.405	.395 (-2.54%)	.447	.439 (-1.81%)	.399	.387 (-2.94%)
Automatic	.422	<u>.374</u> (-11.43%)	.346	<u>.290</u> (-16.39%)	.392	<u>.330</u> (-15.68%)	.423	.369 (-12.88%)	.396	<u>.341</u> (-13.95%)

¹TruFor and CAT-Net were trained on the IMD2020 dataset, which explains their higher performance and robustness to distractions on this dataset.

distractions. Fig. 3 also includes two TruFor examples: the first shows a complete loss of focus on the forgery, while in the second a small part of the forgery is still detected

The observed drop in performance due to forensic distractions highlights the need for distraction-aware IFL models. As TruFor and CAT-Net are both, the best-performing evaluated models, and the most sensitive to distractions, we primarily aim to improve their robustness in Section 4.

4. Proposed Method: Masked Convolutions

This section describes our proposed method of masked convolutions, allowing for distraction-aware forgery localization. Unlike standard convolutions, which convolve the entire image, masked convolutions skip regions covered by a binary mask — treating masked pixels as if they do not exist. By replacing all standard convolutions in the existing IFL model with masked ones, distractions are ignored, allowing the model to refocus on the forgery and improve detection performance. Section 5 implements this approach for TruFor and CAT-Net. However, note that, in theory, this approach is applicable to any CNN-based architecture. Code is available at <https://github.com/IDLabMedia/Masked-convolutions>.

4.1. Masking Components

Distractions introduce artifacts that can interfere with forgery detection. To prevent these artifacts from spreading through the CNN, any component that combines information from multiple pixels (e.g. convolution and pooling layers) must be modified to ignore pixels specified by a binary

mask. Masking a component means altering its operation to ignore masked pixels, effectively treating them as nonexistent. As a reminder, a convolution layer slides a kernel over the input (an image or feature map), at each position computing the dot product between the kernel and the pixels or feature map elements it overlaps with. The output is a new feature map, which is used as input for subsequent layers.

The proposed masked components require an additional input, namely a binary mask indicating the regions to ignore. By blocking the spread of distraction artifacts, the model regains focus on the forgery. The provided mask defines a set of irregularly shaped holes in the image, represented by pixels with a value of 1. All pixels inside the holes are considered *invalid* and should be ignored by the IFL model.

This is achieved in three steps, as illustrated in Fig. 4. (1) Pad: Padding is added to the holes, mirroring the image’s outer padding. This sets the border of the holes to zero while leaving the interior invalid. (2) Convolve: The default convolution operation is performed. (3) Update: Any convolution window that overlaps with non-padded invalid pixels is itself invalid (i.e., its output should be ignored in subsequent layers). The mask is updated to reflect these new validities. Note that applying padding prevents the holes from growing in size after the updates in every layer of the network, while still ensuring that the information within the holes remains isolated throughout inference.

Masking other components follows the same principle. E.g., for pooling layers, this means applying the operation as usual, but only considering non-masked pixels.

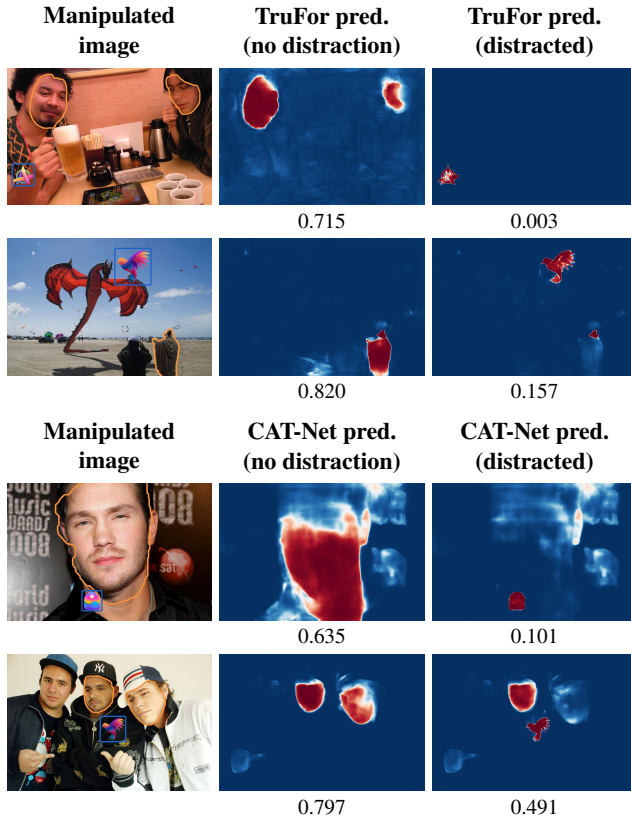


Figure 3. Manipulated images (forgery in orange, distraction in blue) with predictions and corresponding F1* scores (1) for TruFor [4] and CAT-Net [7], evaluated with and without distractions. When distractions are introduced, the models start losing focus on the actual forgeries, though the degree of impact varies across images. Note that the ground truth mask (orange) of the third image, although unexpected, is not an error.

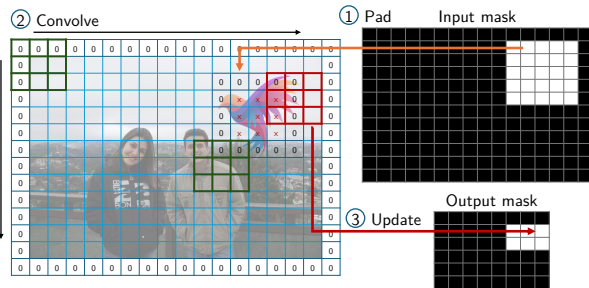


Figure 4. Illustration of the three steps taken by the proposed masked convolution: (1) Pad, (2) Convolve, and (3) Update Mask. For visualization purposes, each block in the grid represents a single pixel or feature map element, and the mask consists of one rectangular hole. The diagram shows a 12×16 input with a 3×3 kernel, stride of 2, and padding of 1. Three example convolution windows are shown: two are valid (green) and one is invalid (red), as it includes invalid pixels from the center of the masked region.

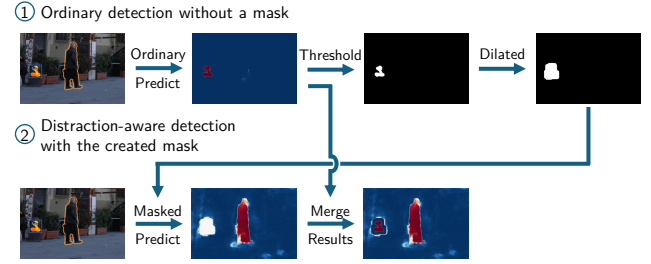


Figure 5. Illustration of the two-step detection process used for automatic distraction-aware FL: (1) Ordinary detection without masking, followed by mask generation via thresholding and dilation. (2) Distraction-aware detection using the generated mask. The final output is created by merging both predictions.

4.2. Integration of Masked Components

Integrating the proposed masked components into a CNN-based model, involves three steps. First, all components that mix information from different pixels are replaced with their masked counterparts. Second, a *distraction mask*, indicating the areas to ignore, is passed through the network alongside the input. Third, each time a masked component is called, the current mask is provided, and the updated mask is captured for use in subsequent layers.

4.3. Distraction Mask Generation

The distraction mask is a binary map with the same height and width as the input image. Pixels marked by the mask will be ignored during inference. We propose two methods for generating masks.

Manual masking A mask can be created manually, for example based on the image under investigation or on previous undesired detection outputs. This approach gives the end user flexibility to exclude specific regions and makes it straightforward to ignore obvious distractions such as logos or overlaid text.

Automatic masking Manually creating a mask for every image is not scalable nor practical in automatic IFL. To address this, we propose an automatic mask generation method. We adopt a two-step detection process as shown in Fig. 5. (1) Ordinary detection: First, detection is performed without a mask. Next, a mask is generated by thresholding the output at 0.5 and applying dilation with a kernel size equal to 5% of the largest dimension (width or height). Thresholding ensures that all pixels detected as altered are ignored, while dilation enlarges the mask to guarantee full coverage of the distraction, thereby removing its influence entirely. (2) Distraction-aware detection: Next, detection is repeated using the generated mask. Finally, the outputs of both steps are merged by replacing masked pixels with their corresponding values from the first prediction. This is necessary as masked pixels produce invalid predictions, which cannot be used directly. Basing the distraction mask on the

model’s own predictions is advantageous, as it targets the exact pixels the model initially flagged as altered. These regions have already received attention during the first pass, so masking them allows the model to shift its focus to other areas that may contain overlooked forgeries.

4.4. Advantages

The use of masked convolutions as a technique for distraction-aware IFL has several advantages. First, it allows the model to focus on the relevant parts of an image while ignoring distractions. Moreover, the approach is toggleable: if no mask is provided, the model behaves as if it were unmodified. In addition, masked convolutions require no additional training or fine-tuning; existing pretrained weights can be reused. This makes the method highly practical compared to alternative approaches, such as retraining with masking-based augmentations which is resource-intensive and requires access to the original training data and setup to avoid degrading generalization performance. Furthermore, by generating the distraction masks automatically (as proposed in Section 4.3), the method enables automated distraction-aware IFL without introducing extra steps compared to the original model. Last, masked convolutions can, in theory, be integrated into any CNN-based architecture, making it a versatile solution. This combination of properties makes masked convolutions a promising approach for improving the robustness of IFL models against forensic distractions.

5. Evaluation

To assess the performance and generalizability of the proposed masked-convolution approach, we integrate the masked components into both the TruFor [4] and CAT-Net [7] models and conduct four experiments. First, we assess the effectiveness of masked convolutions, by manually creating a distraction mask for each image based on the known location and size of the added distraction. Second, we demonstrate the potential for automatic distraction-aware IFL by applying the automated masking method explained in Section 4.3. Third, we evaluate the automatic masked-convolution approach on non-distracted images to ensure that it does not degrade performance. This verification is crucial, as the method should remain safe to apply universally, regardless of the presence of distractions. Finally, we analyze the effect of distraction size on the original TruFor and CAT-Net models and their modified versions. Results of the experiments are presented in Sections 5.2 to 5.5.

5.1. Experimental Setup

We evaluate our method using the same datasets, distraction setup and compression as used in Section 3.1.

Table 2. Impact of distraction size on the performance of unmodified TruFor [4] and CAT-Net [7] (Unmod.) vs. their adapted versions using the proposed masked-convolution approach, evaluated with manual (Manual) and automatic (Auto.) masking and measured in F1* score (1). The original models are more sensitive to smaller distractions, while the adapted versions handle them effectively. Performance on larger distractions is slightly reduced but remains substantially better than the unmodified models. Manual masking achieves the highest accuracy, whereas automatic masking offers convenience at a minor performance cost.

		Model	TruFor [4]		CAT-Net [7]	
			orig	distr (& diff ¹)	orig	distr (& diff ¹)
Small Logos	Unmod.	.399	.152 (-61.77%)	.225	.189 (-16.00%)	
	Manual	.399	.394 (-1.09%)	.225	.217 (-3.22%)	
	Auto.	.396	.358 (-9.66%)	.224	.210 (-6.34%)	
Large Logos	Unmod.	.399	.201 (-49.53%)	.225	.193 (-14.11%)	
	Manual	.399	.380 (-4.79%)	.225	.213 (-5.08%)	
	Auto.	.396	.324 (-18.25%)	.224	.209 (-6.59%)	
Average	Unmod.	.399	.177 (-55.65%)	.225	.191 (-15.06%)	
	Manual	.399	.387 (-2.94%)	.225	.215 (-4.15%)	
	Auto.	.396	.341 (-13.95%)	.224	.209 (-6.46%)	

Detection score Performance is measured using the adapted F1 score (see Equation (1)) as defined in Section 3.1. This ensures the model is neither rewarded nor penalized for detecting distractions. In the manual experiment (Section 5.2), we exclude the entire masked area rather than only the distraction itself. This is necessary because the model produces invalid predictions within masked regions, which would otherwise skew the results.

5.2. Manual Masking

Table 1 shows the mean F1* scores of the original and the modified TruFor and CAT-Net models. Distractions decrease TruFor’s and CAT-Net’s performance by 55.65% and 15.06%, respectively. Using manually generated masks, our proposed masked-convolution approach reduces these drops to just 2.94% and 4.15%, nearly restoring original performance on non-distracted images.

Fig. 6 shows four manipulated images. Columns 2 and 3 display TruFor’s and CAT-Net’s predictions on the original and distracted images, while columns 4 and 5 show the used masks and the predictions of the modified models on the distracted images. Our approach recovers a significant amount of focus, enabling the models to again detect the forgeries. Fig. 2b shows the scatter plot of modified TruFor with manual masking. Points cluster near the diagonal, indicating little sensitivity to distractions.

These results demonstrate the effectiveness of masked convolutions for distraction-aware forgery localization.

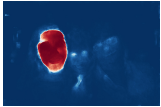
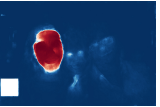
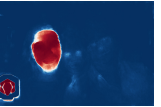
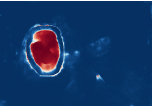
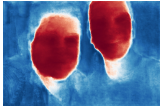
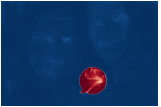
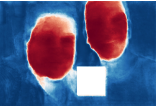
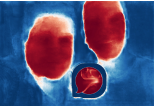
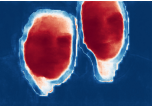
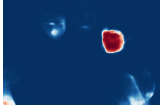
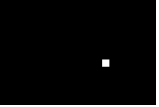
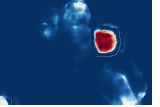
Manipulated image	TruFor pred. (no distraction)	TruFor pred. (distracted)	Mod. TruFor (ours) manual masking (distracted)		Mod. TruFor (ours) automatic masking (distracted)		Mod. TruFor (ours) automatic masking (no distraction)
			manual mask	prediction	automatic mask	prediction	
	 0.833	 0.018		 0.832		 0.817	 0.828
	 0.772	 0.035		 0.839		 0.832	 0.848
Manipulated image	CAT-Net pred. (no distraction)	CAT-Net pred. (distracted)	Mod. CAT-Net (ours) manual masking (distracted)		Mod. CAT-Net (ours) automatic masking (distracted)		Mod. CAT-Net (ours) automatic masking (no distraction)
			manual mask	prediction	automatic mask	prediction	
	 0.793	 0.112		 0.621		 0.481	 0.755
	 0.684	 0.040		 0.671		 0.505	 0.584

Figure 6. Manipulated images (forgery in orange, distraction in blue) with predictions and corresponding F1* scores (1) for TruFor [4], CAT-Net [7], and our proposed modified versions, using both manual masking and automatic masking, evaluated with and without distractions. The modified models significantly outperform the originals in the presence of distractions, demonstrating the effectiveness of the masked-convolution approach. Furthermore, on non-distracted images, the modified models with automatic masking achieve performance comparable to the originals, indicating that the approach can be safely applied in both scenarios.

5.3. Automatic Masking

The automatic masked-convolution approach reduces the performance drop caused by distractions from 55.65% to 13.95% for TruFor and from 15.06% to 6.46% for CAT-Net, as shown in Table 1.

In Fig. 6, columns 6 and 7 show the automatically generated masks and the predictions of the modified models on the distracted images. Despite the imperfect masks, our approach still recovers substantial focus on the forgery, significantly improving robustness compared to the original models. Fig. 2b shows the scatter plot of modified TruFor with automatic masking. Compared to the original model (Fig. 2a), points lie much closer to the diagonal, indicating substantially higher resilience to distractions.

5.4. Automatic Masking on Non-Distracted Images

The bottom part of Table 1 shows the performance of the modified TruFor and CAT-Net models, using automatic masking, on non-distracted images (columns *orig*). Compared to the original models (rows 6 and 7), the average F1* scores show negligible differences (0.399 vs. 0.396 for Tru-

For and 0.225 vs. 0.224 for CAT-Net). These results indicate that the automatic masked-convolution approach can be safely applied to both distracted and non-distracted images, improving robustness when distractions are present while preserving original performance otherwise.

In Fig. 6, column 8 shows the predictions of the modified models using automatic masking on the original (non-distracted) images. These examples illustrate how the approach generally does not negatively impact forgery localization when distractions are absent. The last example shows a slight reduction in detection performance, as the prediction contains some additional background noise.

5.5. Effect of Distraction Size

Table 2 compares original, manually masked, and automatically masked performances for small (between 4–12% of the image width or 0.24–2.16% of the image area) versus large distractions (between 12–24% of the image width or 2.16–8.64% of the image area). TruFor and CAT-Net are both slightly more sensitive to smaller distractions than larger ones, as shown by the difference in performance drop

(61.77% vs. 49.53% for TruFor and 16.00% vs 14.11% for CAT-Net). In contrast, our masked-convolution approach is more effective at masking smaller areas, demonstrated by the manual masking results (1.09% vs. 4.79% for TruFor and 3.22% vs 5.08% for CAT-Net). This trend is even stronger for automatic masking. As distractions increase in size the thresholding-based method sometimes leaves parts uncovered that are too large to be corrected even with dilation, allowing them to remain visible and keep distracting the IFL model.

In summary, masked convolutions prove highly effective in mitigating forensic distractions. Manual masking offers top accuracy but needs user input, while automatic masking is more scalable at the cost of reduced performance. Future work could aim to improve mask generation to narrow this gap. Lastly, the approach generalizes to multiple CNN-based IFL methods, as evidenced by consistent improvements on both TruFor and CAT-Net.

6. Discussion & Future Work

This section discusses the limitations, advantages, and future work opportunities of our proposed distraction-aware approach.

Manually vs. automatically generating masks introduces a trade-off between performance and complexity. Although our proposed two-step automatic approach offers a practical balance, future research could focus on developing more accurate and efficient mask generation techniques to further close the gap between manual and automatic masking.

The observed sensitivity to distractions may highlight a broader limitation: although IFL models can typically detect multiple forgeries within a single image, they may struggle when manipulations are applied at different stages of the editing pipeline. In such cases, one manipulation could interfere with the detection of another. The automatic distraction-aware approach presented in this work could improve performance in such scenarios.

In this paper, we adapted both TruFor and CAT-Net with our approach, chosen for their strong performance on non-distracted images and moderate to high sensitivity to distractions. However, in theory, any CNN-based IFL method can be adapted using masked convolutions.

Finally, the concept of forensic distractions could be extended beyond explicit additions like logos or text overlays, to include attention-drawing regions such as overexposed areas or strong textures that may mislead IFL models.

In summary, we propose that future work focuses on improving the mask generation technique, implementing and evaluating the approach on other sensitive IFL models, addressing robustness under multiple manipulations, and broadening the definition of distractions.

7. Conclusion

In this paper we study the destructive impact of forensic distractions on IFL models and propose a solution to improve their robustness. Our experiments show that forensic distractions can significantly degrade the performance of IFL models. TruFor, in particular, is found to be highly sensitive, causing an average robustness decrease of 55.65%. PSCC-Net and CAT-Net are affected, albeit to a lesser extent, and the evaluated model-based methods show negligible sensitivity.

To address this issue, we introduce masked convolutions, which restrict convolution operations to regions not covered by a provided mask. By replacing all standard convolutions with masked convolutions, the influence of the covered regions should be eliminated, allowing the model to focus on the rest of the image. Additionally, we propose a two-step detection process using thresholding-based masking, enabling automated distraction-aware IFL. A key advantage of this approach is that it does not require re-training, allowing existing models to be adapted seamlessly without modifying their original weights.

Evaluation on TruFor and CAT-Net demonstrates the ability of the masked convolutions to ignore specified regions and its generalizability to different architectures. With manually created masks, the masked convolutions reduce the performance drop from 55.65% to 2.94% for TruFor and from 15.06% to 4.15% for CAT-Net. The proposed automatic masked-convolution approach trades some accuracy for convenience, reducing the performance drop to 13.95% and 6.46%, for TruFor and CAT-Net respectively.

In conclusion, this work contributes to the development of robust multimedia forensics tools by exposing and addressing the challenge of forensic distractions. The proposed approach improves detection performance in distraction-rich environments such as social media, blogs, and news articles.

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